

Migrant vs. Non-Migrant Student Graduation Rates

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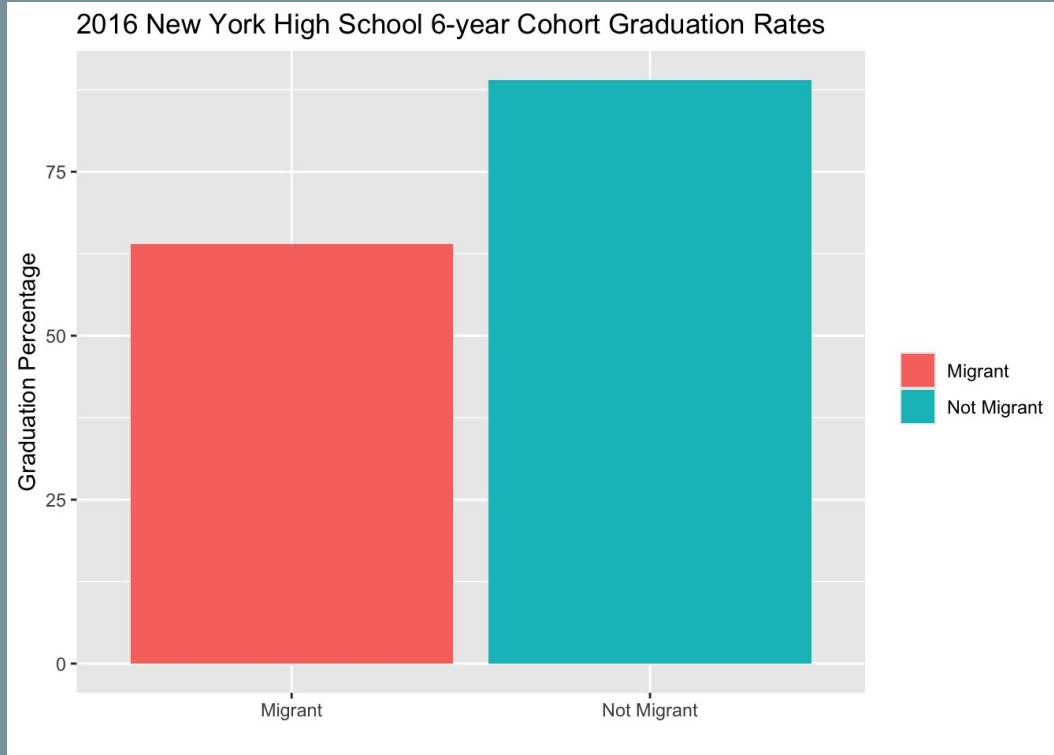
Data Description

- Graduation results for various New York high school populations
 - Analyzed migrant vs non migrant populations graduation rates
- Data downloaded from New York State Education Department
- Have data from 2016-2022, only looking at six-year cohort from 2016-2022

Background

- New York State Education Department (NYSED), which tracks graduation and dropout rates annually across the state
- Key variables included in the dataset that are important to our analysis are migrant status, graduation rate, and dropout rate.
- We will identify the percentage of students who graduated vs those who did not and their migrant status
- For some years there is not data for both migrant and non-migrant graduates.

Visualizations and Summary Statistics



Summary Statistics:

P-value: $4.358553e-16$

Very low p-value signifies that there is a big difference in the proportion of migrant vs non-migrant graduates for the New York High School 6-year cohort.

Discussion

- After conducting our hypothesis test, we are able to determine that there is a significant difference in the graduation rates between migrant and non-migrant students in the 6-year cohort NYSED class of 2022.
- Significant statistical proof that the lives of migrant students can lead to different graduation rates from non-migrant students in the 6-year cohort NYSED class of 2022.
- Shortcomings:
 - There are much more migrant students all over the country and not just in New York. Would we have come to the same conclusion if this was tested on a much larger scale?
 - It would be interesting to see this hypothesis tested with students in similar life situations for more accurate results.
- Assumptions not met:
 - The test we conducted technically does not fit the binomial assumption that each student should have the same probability of graduating. The “S” assumption is not met because there are a lot of different factors that would've affected a student's graduation probability.
 - A deciding factor in the above assumption can be fellow classmates, i.e. the "I" factor, or independence.

Thank You